Risk-Aware Predictive Vertical Handover for UAV Communications in Heterogeneous Wireless Networks

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Abstract

Unmanned aerial vehicles (UAVs) operating in heterogeneous wireless networks require seamless connectivity across multiple radio access technologies, including fifth-generation (5G), Long-Term Evolution (LTE), and Wi-Fi. During aerial missions, UAVs frequently traverse coverage boundaries among these networks, creating vertical-handover (VHO) challenges involving decision latency, ping-pong handovers, and fixed attribute weights that cannot adapt to dynamic channel conditions and flight phases. This study proposes a set of decision-level network selection algorithms to optimize the handover strategy, including the long short-term memory (LSTM)-attention-based adaptive vertical handover algorithm (LAAVHA), adaptive dynamic hysteresis (ADH), and a risk-sensitive technique for order preference by similarity to the ideal solution (RS-TOPSIS), which collectively form the attention-LSTM enhanced risk-aware vertical handover algorithm (ALERA). LAAVHA integrates stacked-LSTM state prediction, attention-based dynamic weighting, improved TOPSIS ranking, and dual-hysteresis decision-making. ADH adapts hysteresis parameters to flight speed and serving-network chan-

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nel fluctuations, while RS-TOPSIS penalizes volatile closeness scores to improve candidate ranking. Simulation results demonstrate reduced unnecessary handovers, improved decision efficiency, sustained remote-sensing data continuity, and greater robustness to abrupt channel variations.

Keywords: UAV communications, heterogeneous wireless networks, vertical handover decision, network selection, LSTM, attention mechanism

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1. Introduction

Unmanned aerial vehicles (UAVs) are increasingly deployed in heterogeneous wireless network environments, where multiple radio access technologies—including fifth-generation (5G), Long-Term Evolution (LTE), and Wi-Fi—coexist with overlapping coverage areas. UAV mobility introduces unique vertical handover (VHO) challenges that differ fundamentally from ground-based scenarios: rapid altitude changes, high-speed boundary crossings, and three-dimensional trajectory variations create dynamic channel conditions that demand intelligent network selection strategies. The core technical challenge lies in managing seamless connectivity across coverage boundaries while maintaining quality of service for real-time aerial data transmission [1, 2, 3]. This study addresses the network selection decision problem: determining when to initiate a handover and which target network to select, based on predicted channel conditions and dynamic attribute weighting. Existing VHO methods mainly employ multiple attribute decision making (MADM) [7], including the technique for order preference by similarity to the ideal solution (TOPSIS) [8, 9, 10], VIsekriterijumska Optimizacija I Kompromisno Resenje (VIKOR) [11], and the Grey relational analysis (GRA), complex proportional assessment (COPRAS), stable preference ordering toward ideal solution (SPOTIS), and simple additive weighting (SAW) methods [12]. Beyond deterministic MADM frameworks, fuzzy-logic-based approaches such as Fuzzy-VHO apply fuzzy inference to handle uncertain channel measurements and handover decisions [4]. More recently, deep reinforcement learning (DRL) methods have been explored to learn adaptive handover policies from experience [16, 17, 18]. Despite their different formulations, these approaches have common limitations. MADM and fuzzy methods typically use fixed weights—often derived from entropy weighting or the analytic hierarchy process [12, 13, 14, 15]—limiting adaptation to varying flight phases and channel conditions, and rely on current network

indicators without predicting future states, potentially delaying handovers until link failure when a UAV approaches coverage limits [16, 17]. DRL-based methods remain limited by black-box decision-making and unstable training, which undermine interpretability and reliability in safety-critical aerial missions.

LSTM networks alleviate vanishing gradients through gated structures [19] and capture long-range temporal dependencies; they have accordingly been used to forecast handover-related network states in 5G heterogeneous networks [20, 21, 22]. Because such forecasting is trained by supervised regression against measured next-period states, it can be inserted into a multi-attribute decision framework as an explicit prediction stage, without replacing the decision rule by a learned policy. In existing schemes, however, the forecast serves mainly to refresh the indicator values that enter the decision layer, whereas the weights and thresholds of that layer are still specified offline; the benefit is therefore confined to smoother inputs rather than adaptive decision logic.

VHO in UAV heterogeneous wireless networks involves two challenges. The first is informational. Conventional MADM ranks candidates on instantaneous indicators and therefore cannot anticipate a link that is about to fail, and although a single LSTM can forecast the next-period state, existing designs use that forecast only to refresh the indicators while leaving the decision rule untouched: the forecast changes what is known, not how the decision is made. The second is parametric. Attribute weights are fixed using entropy weighting or the analytic hierarchy process [12, 13, 14, 15], and the score margin and confirmation window are held constant, yet the relative importance of SINR, delay, throughput, and packet loss differs between cruise and imaging phases, and the channel is quiescent in steady flight but fluctuates sharply near coverage boundaries and obstacles. A margin lax enough to avoid chattering in steady flight responds to boundary noise, while one strict enough to suppress that noise delays necessary handovers, and the same tension governs the length of the confirmation window. Handover instability therefore stems less from limited sensing than from decision logic whose parameters cannot follow the flight state. Addressing these challenges requires a framework that integrates temporal prediction with adaptive decision parameters.

To address these challenges, this paper proposes LAAVHA, which integrates LSTM prediction, attention-based dynamic weighting, improved TOPSIS with fused states, and dual hysteresis. The stacked LSTM addresses the informational challenge by predicting short-term network-state changes;

attention-based weighting addresses the parametric challenge by adapting attribute importance to flight speed and altitude; improved TOPSIS fuses current and predicted states to enable proactive decisions. LAAVHA is further enhanced with ADH and RS-TOPSIS to form ALERA. ADH adapts the handover threshold and confirmation window to serving-network SINR variation and flight speed, addressing the fixed-parameter limitation under varying channel conditions. RS-TOPSIS penalizes historically volatile candidates to prevent false handovers caused by transient score peaks.

Simulation results across 250 extended 50-second trajectories show that LAAVHA achieves the lowest mean handover count (2.13) among 11 algorithms, reducing handovers by $46.0\% - 94.8\%$ versus eight baselines. End-to-end QoS evaluation across 50 representative scenarios demonstrates that LAAVHA simultaneously achieves the highest mean throughput (2.00 Mbps) and lowest mean delay (1.00 ms) among evaluated algorithms. Ablation experiments using 250 seeds with 50-second missions show that removing the LSTM prediction module increases the mean count to 5.55, while removing attention-based dynamic weighting increases it to 24.25, confirming their complementary roles in handover stability. ALERA completes all necessary handovers during sustained link degradation while producing no false handovers under transient fluctuations, whereas a fixed low-threshold scheme causes two additional ping-pong handovers. These results demonstrate that ADH and RS-TOPSIS enhance robustness under channel fluctuations without suppressing necessary handovers.

2. Network Scenario and Network-State Parameters

A UAV aerial mission is considered over a $2000\ \text{m} \times 2000\ \text{m}$ heterogeneous wireless network coverage area, with a UAV following a predefined route at $50 - 150$ m altitude and $5 - 30$ m/s cruise speed. Wi-Fi, LTE, and 5G serve as candidate networks with different coverage ranges and deployment densities. Wi-Fi hotspots have an approximate 200 m radius and are placed near ground control stations and key operational sites. LTE base stations provide $1 - 2$ km coverage and are sparsely deployed along the route. 5G base stations, with an approximate 500 m radius, are concentrated in the eastern data-aggregation area, offering high-rate access but incomplete coverage continuity. The UAV periodically collects the state of each candidate network for handover decisions [23].

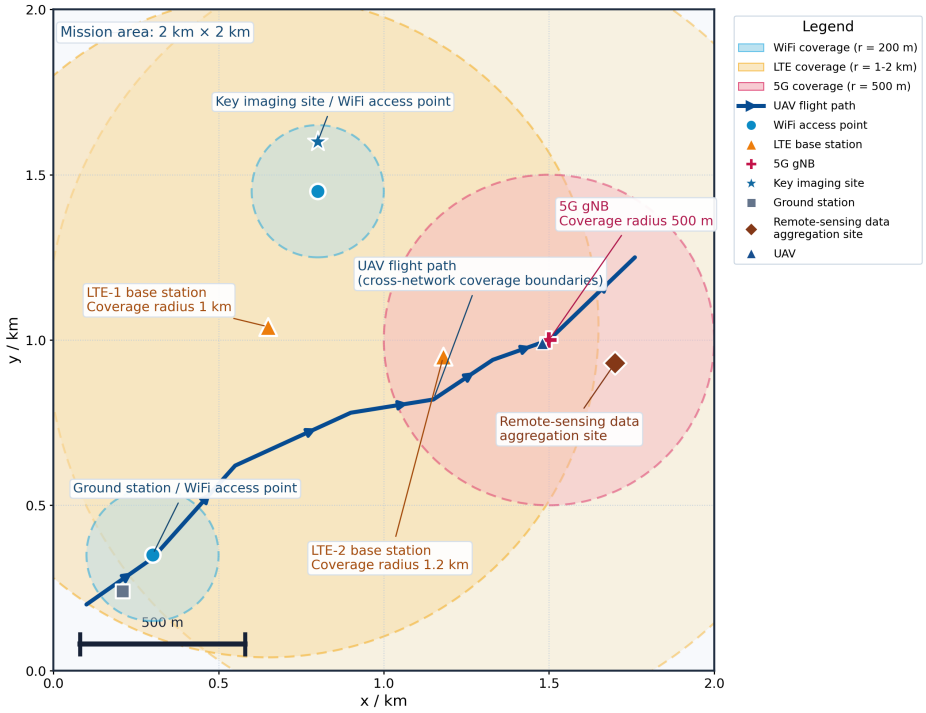


Figure 1: Heterogeneous wireless network coverage and UAV trajectory in the mission area.

Wi-Fi, LTE, and 5G provide overlapping coverage at different scales (Figure 1). Wi-Fi hotspots mainly serve ground stations and key imaging sites; LTE base stations offer relatively continuous wide-area connectivity; and 5G base stations provide high-rate transmission within the eastern data-aggregation area but with limited coverage continuity. The 5G next generation Node B (gNB) provides 5G coverage, radio-resource scheduling, and high-rate data transmission. As the UAV follows its route, it traverses overlapping coverage areas and requires VHO when the serving network degrades; another candidate offers better overall performance. The coverage scales correspond to the specified radii and define the basis for subsequent network-state collection and candidate evaluation.

For candidate network i, the state vector at time t is defined as:

$$S_i(t) = [\mathrm{SINR}_i(t),\ \mathrm{RSRP}_i(t),\ D_i(t),\ T_i(t),\ \mathrm{PLR}_i(t)]\,, \tag{1}$$

where $\mathrm{SINR}_i(t)$, $\mathrm{RSRP}_i(t)$, $D_i(t)$, $T_i(t)$, and $\mathrm{PLR}_i(t)$ denote the signal-to-

interference-plus-noise ratio (SINR), reference signal received power (RSRP), end-to-end delay, throughput, and packet loss ratio (PLR) of candidate i at time t, respectively. These five indicators characterize candidate-network performance from complementary perspectives. As the UAV moves across coverage boundaries, $\mathrm{SINR}_i(t)$ typically declines from tens of decibels toward the noise floor, reflecting signal degradation caused by distance and obstruction and providing a direct handover cue. $\mathrm{RSRP}_i(t)$, together with $\mathrm{SINR}_i(t)$, determines whether the link budget meets the demodulation threshold; below receiver sensitivity, reliable physical-layer connectivity cannot be established even with acceptable SINR. The delay $D_i(t)$ represents the round-trip time of a packet between the ground station and UAV application layer, affecting responsiveness and safety margins during high-speed cruise. Throughput $T_i(t)$ measures deliverable payload bits per unit time; insufficient throughput prevents time-sensitive data payloads from being transmitted during a single overflight. While $\mathrm{PLR}_i(t)$ reflects transmission reliability, even with adequate signal strength and throughput, high $\mathrm{PLR}_i(t)$ can cause data loss or quality degradation. $\mathrm{SINR}_i(t)$, $\mathrm{RSRP}_i(t)$, and $T_i(t)$ are benefit criteria, whereas $D_i(t)$ and $\mathrm{PLR}_i(t)$ are cost criteria. To enable unified multi-attribute evaluation across indicators with different units and ranges, min-max normalization maps all raw measurements to $[0, 1]$:

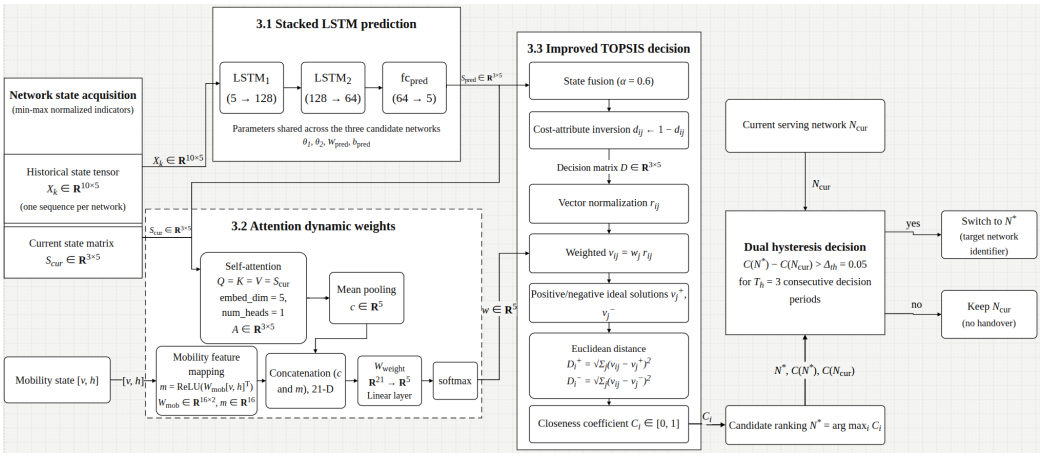
$$s_{\mathrm{norm}} = \frac{s - s_{\min}}{s_{\max} - s_{\min}}, \quad \text{for each indicator } j, \tag{2}$$

where s denotes the raw measurement, and $s_{\min}$ and $s_{\max}$ represent the minimum and maximum values observed across the three candidates at the current decision period. After normalization, larger values consistently indicate better performance, directly supporting benefit criteria (SINR, RSRP, throughput) while requiring subsequent inversion for cost criteria (delay, PLR) in the decision stage [11].

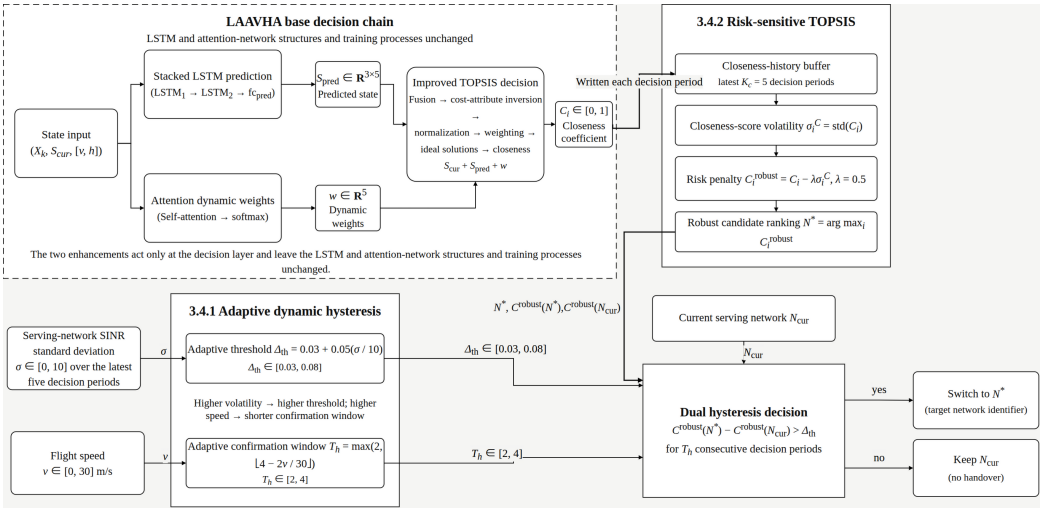
Equations (1) and (2) together define the normalized state representation used throughout the algorithm. Stacking $S_i(t)$ for the three candidates forms the current-state matrix $S_{\mathrm{cur}} \in \mathbf{R}^{3\times5}$ for attention-based weight generation (Section 3.2), while stacking one candidate's states over 10 consecutive periods forms the historical tensor $X_k \in \mathbf{R}^{10\times5}$ for LSTM prediction (Section 3.1). These representations capture both cross-candidate comparison and per-candidate temporal evolution, providing the foundation for the adaptive handover framework in Section 3.

3. LAAVHA Adaptive Vertical Handover Algorithm

Based on the network scenario and network-state parameters (Section 2), this section proposes LAAVHA, an adaptive VHO algorithm that integrates LSTM and an attention mechanism. It overcomes the fixed weighting and lack of temporal prediction in conventional MADM methods for UAV aerial missions in heterogeneous wireless networks. Using the five-dimensional state vector and sliding window (Section 2), LAAVHA performs stacked-LSTM prediction, attention-based dynamic weighting, and improved TOPSIS decision-making with predicted states. ADH and RS-TOPSIS are then incorporated to form ALERA.



(a) LAAVHA algorithm framework



(b) ALERA algorithm framework

Figure 2: LAAVHA and ALERA algorithm frameworks.

LAAVHA comprises network-state prediction, dynamic-weight generation, candidate ranking, and dual-hysteresis decision-making, with ADH and RS-TOPSIS extending the decision chain (Figure 2). These modules form a sequential flow around a common network state. The historical sequence X_k, collected and normalized in Section 2, enters the stacked LSTM to produce the next-period predicted state S_{pred}. Additionally, the current-state matrix S_{cur} enters the attention module with mobility features, including speed and

altitude, to generate dynamic weight w. Then S_{cur}, S_{pred}, and w are combined in improved TOPSIS to obtain candidate closeness score C_i. Finally, C_i and the current serving network N_{cur} enter the hysteresis decision module with ALERA enhancements to determine the final handover. Therefore, Section 3.1 predicts future states, Section 3.2 determines attribute importance, Section 3.3 ranks candidates using current and future states, and Section 3.4 determines whether handover is appropriate under channel fluctuations.

3.1. Network-State Prediction using a Stacked LSTM

The time-window length is set to $T_s = 10$, covering 10 consecutive decision periods over 1.0 s with a decision interval of $\tau = 0.1$ s. Candidate networks $k \in \{1, 2, 3\}$ represent 5G, LTE, and Wi-Fi, respectively. The historical input tensor $X_k \in \mathbf{R}^{10\times 5}$ is constructed by row-wise stacking the normalized five-dimensional state vectors across the 10 periods. The prediction target is the next-period state vector $\hat{s}_k(t+\tau) \in \mathbf{R}^5$:

$$h_{1,k} = \text{LSTM}_1(X_k; \theta_1), \qquad h_{1,k} \in \mathbf{R}^{10\times 128}, \tag{3}$$

$$h_{2,k} = \text{LSTM}_2(h_{1,k}; \theta_2), \qquad h_{2,k} \in \mathbf{R}^{10\times 64}, \tag{4}$$

$$\hat{s}_k(t+\tau) = W_{\text{pred}} h_{2,k}[-1] + b_{\text{pred}}, \qquad \hat{s}_k \in \mathbf{R}^5. \tag{5}$$

where θ_1 and θ_2 denote the trainable parameters of the two LSTM layers, including the weights and biases of the input, forget, and output gates and cell states, while $W_{\text{pred}} \in \mathbf{R}^{5\times 64}$ and $b_{\text{pred}} \in \mathbf{R}^5$ represent the fully connected prediction-head weight and bias, respectively, and $h_{2,k}[-1]$ indicates the final output of LSTM_2. Equations (3)–(5) define the prediction pipeline: 10 steps $\times$ 5 indicators $\rightarrow \text{LSTM}_1(5 \rightarrow 128) \rightarrow \text{LSTM}_2(128 \rightarrow 64) \rightarrow \text{fc}_{\text{pred}}(64 \rightarrow 5)$. The three candidate networks share θ_1, θ_2, W_{pred}, and b_{pred}, while using their respective historical sequences X_k. The resulting predictions are stacked into $S_{\text{pred}} \in \mathbf{R}^{3\times 5}$, representing the next-period values of the five indicators for each network. The model is trained offline using supervised learning with mean squared error (MSE) loss:

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_{n=1}^{N} \left\| S_{\text{pred}}^{(n)} - S_{\text{target}}^{(n)} \right\|_2^2, \tag{6}$$

where $n = 1, \ldots, N$ indexes training samples, each comprising a historical sequence $X_k^{(n)}$ and the corresponding ground-truth next-period state $S_{\text{target}}^{(n)}$. Training hyperparameters are detailed in Section 4.1. The predicted state S_{pred} is subsequently fused with the current state S_{cur} in the TOPSIS decision stage (Section 3.3), enabling forward-looking candidate evaluation.

3.2. Dynamic Weight Generation for Flight Phases

Network-attribute importance varies with the VHO decisions. During cruise, link stability and flight-control delay are more critical to safety, whereas during image return at imaging sites, throughput and packet loss become more important for timely and reliable transmission of high-resolution images [24]. Fixed weights cannot capture these changing priorities.

To capture phase-dependent requirements, LAAVHA employs self-attention [26, 27]. The five-dimensional state vectors of the three candidates are row-wise stacked into $S_{\text{cur}} \in \mathbf{R}^{3\times5}$, with $Q = K = V = S_{\text{cur}}$, `embed_dim` $= 5$, and `num_heads` $= 1$. The `embed_dim` denotes the feature dimension of each attention input and is set to 5, corresponding to the normalized SINR, RSRP, delay, throughput, and packet loss ratio indicators, and `num_heads` represents the number of parallel attention heads and is set to 1. With five attributes, multiple heads would provide too few features per head and reduce weight interpretability. A single head therefore captures attribute correlations directly. The self-attention mechanism captures cross-network dependencies among the candidate networks, where the attention matrix quantifies how each network's state should be aggregated based on inter-network relationships. Cross-network mean pooling then extracts the attribute-level context $c \in \mathbf{R}^5$ from the attention output. The resulting representations are combined with UAV mobility features to generate dynamic attribute weights for subsequent TOPSIS-based network selection. Flight speed v and altitude h are mapped by $W_{\text{mob}} \in \mathbf{R}^{16\times2}$ to mobility feature $m \in \mathbf{R}^{16}$. The concatenated results are projected by a fully connected layer and normalized by softmax to obtain the five-dimensional dynamic weight w:

$$Q = K = V = S_{\text{cur}} \in \mathbf{R}^{3\times5}, \tag{7}$$

$$A = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{5}}\right)V, \quad A \in \mathbf{R}^{3\times5}, \tag{8}$$

$$c = \text{mean}(A, \text{dim} = 0) \in \mathbf{R}^5, \tag{9}$$

$$m = \text{ReLU}\big(W_{\text{mob}}[v, h]^\top\big), \quad m \in \mathbf{R}^{16}, \tag{10}$$

$$w = \text{softmax}(W_{\text{weight}}[c \parallel m]), \quad W_{\text{weight}} : \mathbf{R}^{21} \to \mathbf{R}^5, \quad w \in \mathbf{R}^5, \quad \sum_j w_j = 1. \tag{11}$$

Equations (7)–(11) define the dynamic-weight procedure: $S_{\text{cur}} \in \mathbf{R}^{3\times5}$ enters self-attention, producing the 3×5 matrix A; cross-network mean pooling yields the five-dimensional attribute context c; speed and altitude are

mapped to the 16-dimensional mobility feature m; and their 21-dimensional concatenation is projected by W_{weight} and softmax-normalized to obtain the five-dimensional dynamic weight w. The dynamic weight w is subsequently applied in the TOPSIS ranking stage (Section 3.3), enabling SINR, RSRP, delay, throughput, and packet loss ratio to contribute adaptively to the final closeness score C_i while the LSTM branch predicts future state changes.

3.3. Improved TOPSIS Decision Making with Fused Predicted States

Since aerial mission performance depends jointly on signal strength, transmission capability, and reliability, no single indicator defines the optimal network. TOPSIS ranks candidates by their proximity to the virtual ideal and negative-ideal networks [8, 9]. To incorporate future conditions, LAAVHA fuses the current state S_{cur} and predicted state S_{pred} to form the decision matrix:

$$d_{ij} = \alpha s_{\text{cur}}(i,j) + (1-\alpha) s_{\text{pred}}(i,j), \quad \alpha = 0.6, \tag{12}$$

$$d_{ij} \leftarrow 1 - d_{ij}, \quad j \in \{\text{delay}, \text{PLR}\}. \tag{13}$$

where $\alpha = 0.6$ (justified in Section 4.1) balances current reliability with predictive foresight, and Equation (13) inverts delay and PLR so all attributes follow larger-is-better orientation. The resulting decision matrix $D \in \mathbf{R}^{3\times 5}$ enters standard TOPSIS:

$$r_{ij} = \frac{d_{ij}}{\sqrt{\sum_{k=1}^{3} d_{kj}^2}}, \tag{14}$$

$$v_{ij} = w_j r_{ij}, \tag{15}$$

$$v_j^+ = \max_i v_{ij}, \quad v_j^- = \min_i v_{ij}, \tag{16}$$

$$D_i^+ = \sqrt{\sum_{j=1}^{5} (v_{ij} - v_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^{5} (v_{ij} - v_j^-)^2}, \tag{17}$$

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}, \quad C_i \in [0,1]. \tag{18}$$

where r_{ij} denotes the vector-normalized element of fused matrix D, and $\sqrt{\sum_{k=1}^{3} d_{kj}^2}$ represents the Euclidean norm of attribute j across the three candidates, with k as the summation index, w_j indicates the dynamic weight

of attribute j from the attention module, v_{ij} signifies the weighted normalized element, v_j^+ and v_j^- correspond to the positive and negative ideal solutions, defined as the largest and smallest weighted normalized values for attribute j, respectively, D_i^+ and D_i^- are the distances of candidate i from these solutions, and $C_i \in [0, 1]$ is its relative closeness. Therefore, Equations (14)–(18) convert the 3×5 fused matrix into three closeness scores through normalization, dynamic weighting, ideal-solution determination, and Euclidean-distance calculation. A value closer to 1 indicates stronger overall support for image transmission, flight-control responsiveness, and link reliability.

$$N^* = \arg\max_i C_i, \tag{19}$$

$$\text{Decision} = \begin{cases} \text{switch to } N^*, & \min_{l=0}^{T_h-1} \left[C_t(N^*) - C_{t-l}(N_{\text{cur}})\right] > \Delta_{\text{th}}, \\ \text{keep } N_{\text{cur}}, & \text{otherwise}, \end{cases} \tag{20}$$

where N^* denotes the candidate with the highest current closeness, N_{cur} represents the serving network, $C_t(\cdot)$ indicates the closeness score at decision period t, $\Delta_{\text{th}} = 0.05$ specifies the required score advantage, and $T_h = 3$ specifies the confirmation window (justified in Section 4.1). The min operator enforces that the advantage must persist across all of the latest T_h periods, suppressing transient-fluctuation-induced ping-pong handovers.

Algorithm 1 implements the three-stage LAAVHA procedure. Stage 1 predicts S_{pred} via stacked LSTM and generates dynamic weight w via attention. Stage 2 fuses S_{cur} and S_{pred} with cost-attribute inversion, applies dynamic weighting, and ranks candidates by TOPSIS closeness C_i. Stage 3 triggers handover only when the advantage persists for T_h consecutive periods, suppressing transient-fluctuation-induced ping-pong handovers.

Algorithm 1 LAAVHA vertical-handover decision procedure

Require: Historical states $\{X_k\}_{k=1}^{3}$, current state S_{cur}, mobility state $[v, h]$, serving network N_{cur}, previous confirmed candidate N_{cand}, and its consecutive count q

Ensure: Handover decision Decision and updated N_{cur}, N_{cand}, and q

Stage 0: Initialization
Read the current decision-period state and previous hysteresis state N_{cand}, q
Stage 1: State prediction and dynamic-weight generation
for $k \leftarrow 1$ **to** 3 **do**
    $h_{1,k} \leftarrow \text{LSTM}_1(X_k; \theta_1)$, $h_{2,k} \leftarrow \text{LSTM}_2(h_{1,k}; \theta_2)$
    $s_{\text{pred},k} \leftarrow W_{\text{pred}} h_{2,k}[-1] + b_{\text{pred}}$
end for
Stack $\{s_{\text{pred},k}\}_{k=1}^{3}$ as S_{pred}
$A \leftarrow \text{Attention}(S_{\text{cur}}, S_{\text{cur}}, S_{\text{cur}})$, $c \leftarrow \text{mean}(A, \text{dim} = 0)$
$m \leftarrow \text{ReLU}(W_{\text{mob}}[v, h]^{\top})$, $w \leftarrow \text{softmax}(W_{\text{weight}}[c \parallel m])$
Stage 2: Fused-TOPSIS candidate ranking
for $i \leftarrow 1$ **to** 3 **do**
    for $j \leftarrow 1$ **to** 5 **do**
        $d_{ij} \leftarrow 0.6 s_{\text{cur}}(i, j) + 0.4 s_{\text{pred}}(i, j)$
        if $j \in \{\text{delay}, \text{PLR}\}$ **then** $d_{ij} \leftarrow 1 - d_{ij}$
        end if
    end for
end for
Compute r_{ij}, v_{ij}, v_j^+, v_j^-, D_i^+, D_i^-, and C_i using Equations (14)–(18)
$N^* \leftarrow \arg\max_i C_i$
Stage 3: Dual-hysteresis handover decision
if $N^* \neq N_{\text{cand}}$ **then**
    $N_{\text{cand}} \leftarrow N^*$, $q \leftarrow 0$
end if
if $C(N^*) - C(N_{\text{cur}}) > \Delta_{\text{th}}$ **then**
    $q \leftarrow q + 1$
else
    $q \leftarrow 0$
end if
if $q \geq T_h$ **then**
    Decision $\leftarrow$ switch to N^*, $N_{\text{cur}} \leftarrow N^*$
else
    Decision $\leftarrow$ keep N_{cur}
end if
return Decision, N_{cur}, N_{cand}, q

3.4. ALERA Enhanced Risk-Aware Vertical Handover Algorithm

The fixed Δ_{th} and T_h in Section 3.3 cannot adapt to varying channel conditions: during coverage-boundary crossings or obstructions, conservative thresholds delay necessary handovers, while low thresholds cause ping-pong handovers under transient fluctuations at high speed. Standard TOPSIS further ranks candidates solely by instantaneous closeness C_i, making decisions vulnerable to candidates with high but volatile scores. ADH addresses the first limitation by adjusting Δ_{th} and T_h from serving-network SINR variation and flight speed, while RS-TOPSIS addresses the second by penalizing closeness-score instability to derive C_i^{robust}. Together they extend LAAVHA into ALERA.

3.4.1. Adaptive Hysteresis Parameters

The fixed $\Delta_{\text{th}} = 0.05$ and $T_h = 3$ (Equation (20)) apply the same handover barrier across all channel conditions and flight phases. Under stable channels at low speed, such conservatism effectively suppresses ping-pong; however, during coverage-boundary crossings or obstructions, it delays necessary handovers, increasing link-interruption risk. Conversely, at high speed, the same window permits excessive spatial displacement before confirmation completes, reducing confirmation effectiveness. Adaptive hysteresis margins and trigger windows have been shown to improve handover decisions in heterogeneous and ultra-dense networks [28, 29, 30, 31]. Accordingly, ADH replaces fixed parameters with context-aware values derived from serving-network SINR variation and flight speed. Let σ denote the standard deviation of serving network SINR over the latest five decision periods, and v denote real-time flight speed. Both are already available: σ is derived from the SINR history in Equation (1), and v is the mobility input used for dynamic weighting in Section 3.2. The adaptive threshold and window are:

$$\Delta_{\text{th}} = 0.03 + 0.05 \left(\frac{\sigma}{10} \right), \quad \sigma \in [0, 10], \tag{21}$$

$$T_h = \max \left(2, \left\lfloor 4 - \frac{2v}{30} \right\rfloor \right), \quad v \in [0, 30] \text{ m/s}. \tag{22}$$

Equation (21) sets a base threshold of 0.03 under stable channels ($\sigma \approx 0$), allowing faster response to sustained degradation, and increases linearly to 0.08 under severe variation ($\sigma \approx 10$), strengthening the handover barrier. Equation (22) decreases the confirmation window from 4 periods at low speed to 2 at high speed, balancing confirmation duration against position change:

at 30 m/s, $T_h = 4$ permits 12 m of movement before confirmation completes, while $T_h = 2$ limits this to 6 m, better supporting timely boundary handover.

3.4.2. Risk-Sensitive TOPSIS

Original TOPSIS ranks candidates by current closeness C_i, obtained from Equation (18) by fusing S_{cur}, S_{pred}, and w. RS-TOPSIS retains this prediction and weighting process while adding a risk constraint to the C_i series. Based on the lower-confidence-bound (LCB) principle, a historical-variation penalty is subtracted from each candidate score:

$$C_i^{\text{robust}} = C_i - \lambda \sigma_i^C, \quad \lambda = 0.5, \tag{23}$$

where σ_i^C denotes the standard deviation of the latest five scores for network i, and $\lambda = 0.5$ is the risk-aversion coefficient (λ setting justified in Section 4.1). When current scores are similar, the mechanism favors the more stable candidate over one with a slightly higher mean but greater variation. For example, if LTE closeness increases steadily with low variance while Wi-Fi closeness fluctuates sharply, C_i^{robust} favors LTE even with slightly higher current Wi-Fi closeness, improving service continuity.

Algorithm 2 enhances the base closeness score at each decision period. Its C_i input comes from Algorithm 1's TOPSIS steps. RS-TOPSIS first computes the risk-penalized ranking, ADH then determines adaptive thresholds, and the dual-hysteresis mechanism finally decides based on persistent robust-score advantage.

Algorithm 2 ALERA risk-aware vertical-handover enhancement procedure

Require: Base closeness $\{C_i\}_{i=1}^{3}$ and its latest $K_c = 5$ history, latest five serving-network SINR values, flight speed v, serving network N_{cur}, previous confirmed candidate N_{cand}, and its consecutive count q

Ensure: Enhanced handover decision Decision and updated N_{cur}, N_{cand}, and q

Stage 1: RS-TOPSIS risk-sensitive ranking
for $i \leftarrow 1$ **to** 3 **do**
    Append C_i to closeness history and keep the latest K_c periods
    $\sigma_i^C \leftarrow \text{std}(\text{closeness history of } C_i)$
    $C_i^{\text{robust}} \leftarrow C_i - 0.5\sigma_i^C$
end for
$N^* \leftarrow \arg\max_i C_i^{\text{robust}}$
Stage 2: ADH adaptive-hysteresis parameter update
$\sigma \leftarrow \text{std}(\text{latest five SINR values of } N_{\text{cur}})$
$\Delta_{\text{th}} \leftarrow 0.03 + 0.05(\sigma/10)$, $T_h \leftarrow \max(2, \lfloor 4 - 2v/30 \rfloor)$
Stage 3: Robust-score dual-hysteresis decision
if $N^* \neq N_{\text{cand}}$ **then**
    $N_{\text{cand}} \leftarrow N^*$, $q \leftarrow 0$
end if
if $C^{\text{robust}}(N^*) - C^{\text{robust}}(N_{\text{cur}}) > \Delta_{\text{th}}$ **then**
    $q \leftarrow q + 1$
else
    $q \leftarrow 0$
end if
if $q \geq T_h$ **then**
    Decision $\leftarrow$ switch to N^*, $N_{\text{cur}} \leftarrow N^*$
else
    Decision $\leftarrow$ keep N_{cur}
end if
return Decision, N_{cur}, N_{cand}, q

3.5. Complexity Analysis and Algorithm Summary

Table 1 summarizes the inputs, outputs, core procedures, and computational complexity of LAAVHA and ALERA. The stacked-LSTM prediction module (Section 3.1) processes the historical state tensor $X_k \in \mathbf{R}^{10\times5}$ through two LSTM layers to generate $S_{\text{pred}} \in \mathbf{R}^{3\times5}$, with complexity $O(R \cdot K \cdot d \cdot h^2)$. The dynamic-attention module (Section 3.2) processes $S_{\text{cur}} \in \mathbf{R}^{3\times5}$, incorporates mobility features, and outputs $w \in \mathbf{R}^5$, with complexity $O(R^2 \cdot d)$. Improved TOPSIS (Section 3.3) fuses and normalizes S_{cur} and S_{pred}, then ranks

candidates using weights, producing C_i with complexity $O(R \cdot d)$. ADH and RS-TOPSIS apply statistical updates to SINR, speed, and C_i history with $O(1)$ and $O(R \cdot K_c)$ complexity, respectively.

Table 1: Summary of the algorithms.

Module	**Input/output**	**Core procedure**	**Complexity**
Stacked-LSTM prediction	In: $3 \times 10 \times 5$ state tensor; out: predicted state	$\text{LSTM}_1(5 \to 128) \to \text{LSTM}_2(128 \to 64) \to \text{fc}(64 \to 5)$	$O(R \cdot K \cdot d \cdot h^2)$
Attention dynamic weights	In: 3×5 state matrix + mobility features; out: 5-D dynamic weights	Self-Attn→mean pooling→ concatenate mobility features→softmax	$O(R^2 \cdot d)$
Improved TOPSIS	In: current + predicted states + dynamic weights; out: candidate closeness C_i	Fused matrix→ weighted normalization→ positive/negative ideal solutions→ closeness	$O(R \cdot d)$
ADH	In: SINR history + flight speed; out: adaptive Δ_{th}, T_h	SINR-variation calculation→linear threshold adjustment→window adaptation	$O(1)$
RS-TOPSIS	In: recent closeness history; out: LCB-adjusted ranking	Historical standard deviation→penalty calculation→ C_i^{robust} ranking	$O(R \cdot K_c)$

4. Simulation Experiments and Results Analysis

4.1. Experimental Platform and Parameter Settings

To evaluate LAAVHA/ALERA in UAV heterogeneous wireless networks, a simulation and inference platform was developed using ns-3.45 [31] and its AI extension ns3-ai [32]. ns-3 implements the 5G/LTE/Wi-Fi topology, UAV mobility, and physical- and transport-layer measurements, while ns3-ai enables bidirectional simulation-inference communication through shared

memory. Each decision period transfers a 10-step history of five indicators for three candidate networks and the mobility state to shared memory. The inference module loads the trained LAAVHA model, reads the network state, speed, altitude, and current-network identifier, and returns the target-network identifier and candidate scores. The model input comprises 3 networks $\times 10$ time steps $\times 5$ indicators (150 dimensions), with each network processed independently by the stacked LSTM. Mobility input comprises speed and altitude. The model outputs a five-dimensional predicted state and dynamic weight, followed by candidate scoring through improved TOPSIS. The testing dataset uses seeds 200–449 (250 trajectories for algorithm comparison; the first 50 seeds, 200–249, are also used for ablation and QoS experiments). Each simulation lasts 50 s with a 0.1 s decision period, yielding 500 decisions per trajectory. Position and altitude perturbations of 30 m and 10 m, respectively, represent flight deviations.

Before finalizing the parameters in Section 3, three sensitivity experiments were conducted to tune the core algorithmic parameters: the fusion coefficient α (balancing current and predicted states), dual-hysteresis thresholds Δ_{th} and T_h (controlling handover triggering), and enhancement parameters K_c and λ (managing risk sensitivity). All experiments used carefully designed stress scenarios that isolate the effect of each parameter group. Figure 3 summarizes the parameter-sensitivity results.

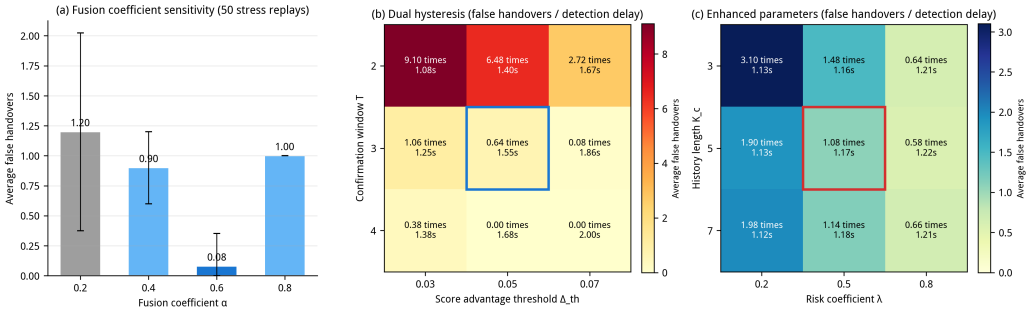


Figure 3: Sensitivity analysis of the fusion coefficient, dual-hysteresis parameters, and enhancement parameters. (For (**b**) and (**c**), upper entries denote mean false-handover counts and lower entries denote mean detection delays.)

(a) Fusion coefficient α. To evaluate the trade-off between current measurements and predicted states, 50 stress replays involving both sustained link degradation and transient channel fluctuations were constructed. Cur-

rent SINR observations captured instantaneous channel-measurement variation, while LSTM predictions exhibited short-term lag under abrupt changes. For $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$, mean false-handover counts were 1.20, 0.90, 0.08, and 1.00, respectively. Lower α (heavier reliance on prediction) increased sensitivity to prediction lag during rapid changes, whereas higher α (heavier reliance on current state) increased sensitivity to measurement noise. The optimal $\alpha = 0.6$ minimized false handovers by assigning 60% weight to current measurements and 40% to predictions.

(b) Dual-hysteresis parameters Δ_{th} and T_h. Using the selected $\alpha = 0.6$, parameters were evaluated on 50 trajectories containing two critical intervals: sustained degradation requiring handover, and transient fluctuations that should not trigger handover. Across $\Delta_{\text{th}} \in \{0.03, 0.05, 0.07\}$ and $T_h \in \{2, 3, 4\}$, all combinations achieved 100% necessary-handover detection. The setting $\Delta_{\text{th}} = 0.05, T_h = 3$ yielded 0.64 mean false handovers and 1.55 s mean detection delay. The aggressive $0.03, 2$ setting reduced delay but produced 9.10 false handovers, while the conservative $0.07, 4$ setting eliminated false handovers at the cost of 2.00 s delay. Therefore, $\Delta_{\text{th}} = 0.05, T_h = 3$ was selected to balance stability and responsiveness.

(c) Enhancement parameters K_c and λ. Using the tuned $\alpha = 0.6$ and $\Delta_{\text{th}} = 0.05, T_h = 3$, the RS-TOPSIS risk-penalty parameters were evaluated on the same 50 trajectories. All combinations of $K_c \in \{3, 5, 7\}$ and $\lambda \in \{0.2, 0.5, 0.8\}$ achieved 100% necessary-handover detection. At $K_c = 5$, increasing λ from 0.2 to 0.8 reduced mean false handovers from 1.90 to 0.58 but increased detection delay from 1.13 s to 1.22 s. At $\lambda = 0.5$, varying K_c from 3 to 7 showed that $K_c = 5$ produced the lowest false-handover count (1.08) with moderate delay (1.17 s). Therefore, $K_c = 5, \lambda = 0.5$ were selected.

Based on these parameter-selection experiments, Table 2 summarizes the platform, network scenario, model architecture, and training/inference parameters. All comparison algorithms use the same simulation configuration, indicator sources, and analysis setting.

Table 2: Experimental parameter settings.

Parameter	**Value**
Simulation platform	ns-3.45, ns3-ai, Python/PyTorch
Candidate networks	5G, LTE, Wi-Fi
Algorithm scope	LAAVHA, 8 comparison algorithms, and 2 ablation variants
Runs	250 trajectories (11-algorithm comparison); 50 (ablation/QoS)
Simulation duration	50 s per trajectory
Decision period	0.1 s
Decisions per run	500
Random seeds	200–449 (11-algorithm comparison); 200–249 (ablation/QoS)
FlowMonitor mode	feed
Indicator dimensions	SINR, RSRP, delay, throughput, packet loss ratio
Position perturbation	30 m
Altitude perturbation	10 m
Time window T_s	10
LSTM hidden units	128 in the first layer, 64 in the second layer
Attention heads (num_heads)	1 (embed_dim=5; single-head attention)
Fusion coefficient α	0.6
Dual-hysteresis parameters	$\Delta_{\text{th}} = 0.05$, confirmation window $T_h = 3$
Enhancement parameters	RS-TOPSIS closeness-history length $K_c = 5$, RS-TOPSIS risk coefficient $\lambda = 0.5$
Training dataset	10,000 samples (historical states and next-period targets)
Dataset split	training/validation = 80%/20% (sample-level random split; see text)
Loss function	MSE (mean squared error) for LSTM prediction only
Optimizer	Adam (learning rate = 0.001)
Batch size	64
Training epochs	50 (maximum)
Early stopping	validation loss does not decrease for 10 consecutive epochs
Trainable parameters	LSTM layers (θ_1, θ_2) and prediction head (W_{pred}, b_{pred})
Fixed parameters	Attention mechanism, W_{mob}, and W_{weight} (Xavier uniform initialization)

In addition to LAAVHA, the experiments evaluate eight comparison algorithms, namely TOPSIS-Q, Fuzzy-VHO, SAW, VIKOR, GRA, COPRAS, SPOTIS, and strongest-signal selection, plus the ablation variants LAAVHA-L and LAAVHA-A. These methods represent seven decision paradigms: fuzzy logic (Fuzzy-VHO), SAW, compromise ranking (VIKOR), distance-based ranking (TOPSIS-Q), relational analysis (GRA), proportional assessment (COPRAS), and reference-point ranking (SPOTIS). TOPSIS-Q follows TOPSIS network selection with entropy-weighted improved TOPSIS [8, 9, 10]; Fuzzy-VHO follows fuzzy handover for 5G UAV networks [4]; VIKOR uses compromise ranking [11]; and GRA, COPRAS, SPOTIS, and SAW represent mainstream multi-attribute mobility decision methods [12]. Strongest-signal selection chooses the candidate with the highest SINR. LAAVHA-L removes LSTM prediction, replacing predicted states with current states while retaining attention weights and dual hysteresis. LAAVHA-A removes attention-based dynamic weighting, retaining LSTM prediction and dual hysteresis while using entropy weighting for TOPSIS.

The LSTM prediction module requires training before deployment. To ensure valid generalization evaluation, the training and testing datasets must use disjoint random seeds. The training dataset uses seeds 0–199 to generate 10,000 supervised samples (historical 10-step windows paired with next-period ground-truth states). Seeds 200–449 are reserved exclusively for testing: all 250 seeds (200–449) are used for the 11-algorithm handover-count comparison and ablation study, while the first 50 of these (200–249) are additionally used for QoS evaluation. This separation prevents temporal information leakage and ensures that all performance metrics reflect the model's ability to generalize to unseen trajectories rather than memorize training scenarios. The training dataset is split 80%/20% at the sample level into training and validation subsets (8,000/2,000 samples). Training hyperparameters (optimizer, learning rate, early stopping, etc.) are detailed in Table 2.

Network-state indicators are collected through ns-3 measurement modules. SINR and RSRP are derived from position-based propagation models [33], while transport indicators (delay, throughput, packet loss ratio) are obtained from FlowMonitor or PacketSink statistics. Because ns-3.45 does not provide a complete 5G NR implementation, the 5G candidate uses a point-to-point proxy link with transport indicators obtained from proxy-flow FlowMonitor. Table 3 summarizes the indicator sources for each candidate network.

Table 3: Candidate-network indicator sources.

Network	**SINR/RSRP**	**Delay**	**Throughput**	**Packet loss ratio**
Wi-Fi	Propagation proxy from MobilityModel position	FlowMonitor	PacketSink bytes received within interval	FlowMonitor
LTE	Propagation proxy from MobilityModel position	FlowMonitor	FlowMonitor	FlowMonitor
5G	Propagation proxy to assumed gNB	Point-to-point proxy-flow FlowMonitor	Point-to-point proxy-flow FlowMonitor	Point-to-point proxy-flow FlowMonitor

4.2. Horizontal Comparison of Algorithms

To evaluate LAAVHA across decision paradigms, eight representative baseline algorithms were tested under identical conditions, covering five MADM variants and three alternative approaches. The MADM baselines include TOPSIS-Q (entropy-weighted TOPSIS without temporal prediction), SAW (simple additive weighting), VIKOR (compromise ranking balancing group utility and individual regret), GRA (grey relational analysis with uncertain information handling), COPRAS (proportional assessment of benefit and cost criteria), and SPOTIS (reference-point ranking). Fuzzy-VHO represents rule-based handover using a Mamdani fuzzy-inference system on SINR and RSRP. Strongest-signal selection serves as a single-attribute baseline that chooses the candidate with the highest SINR. Additionally, two ablation variants are evaluated: LAAVHA-L (without LSTM prediction) and LAAVHA-A (without attention-based dynamic weighting). Figure 4 compares all 11 algorithms.

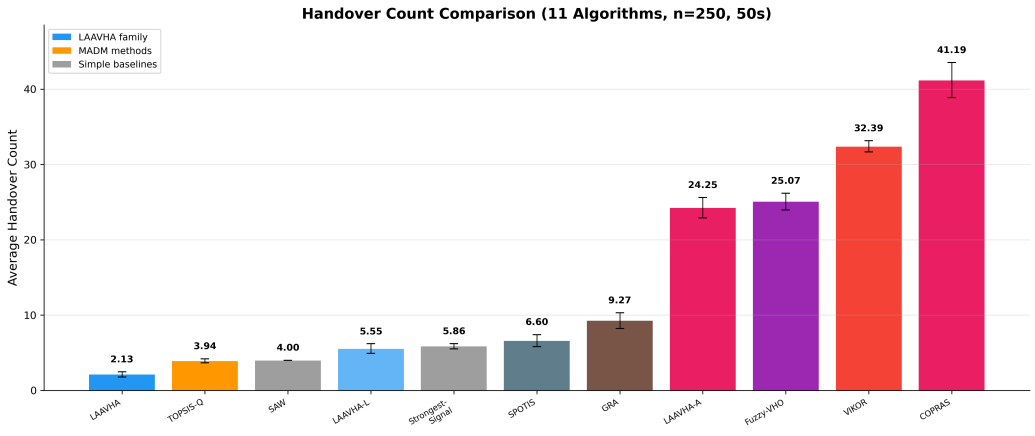


Figure 4: Mean handover-count comparison of 11 algorithms (mean ± std, n=250, 50-second trajectories).

LAAVHA achieves the lowest mean handover count among 11 algorithms, at 2.13 (Figure 4). Based on the heterogeneous coverage topology and UAV trajectory in Figure 1 (Section 2), the UAV traverses from 5G-dominant regions through Wi-Fi hotspots into LTE-covered areas, requiring approximately 2 handovers under ideal decision-making. LAAVHA's mean of 2.13 approaches this ideal baseline. The three LAAVHA variants (LAAVHA 2.13, LAAVHAL 5.55, LAAVHAA 24.25) demonstrate the contributions of individual components, with the full algorithm outperforming ablated versions. Among the baseline algorithms, TOPSIS-Q yields 3.94, SAW 4.00, strongest-signal selection 5.86, SPOTIS 6.60, GRA 9.27, Fuzzy-VHO 25.07, VIKOR 32.39, and COPRAS 41.19. MADM methods without temporal prediction (TOPSIS-Q, SAW, VIKOR, GRA, SPOTIS, COPRAS) exhibit elevated handover counts because fixed weights and instantaneous indicators cause repeated reassignments when candidate scores alternate near coverage boundaries. Strongest-signal selection considers only SINR, excluding delay, throughput, and packet loss. Fuzzy-VHO's rule-based membership functions respond reactively to local SINR and RSRP fluctuations. Overall, LAAVHA reduces handovers by $46.0\% - 94.8\%$ versus the eight baselines, confirming that temporal prediction and adaptive weighting mitigate the limitations observed in fixed-weight and reactive approaches.

To address whether LAAVHA's low handover count stems from intelligent decision-making or excessive conservatism, end-to-end communication quality is evaluated across five representative algorithms under 50 extended-

duration scenarios (seeds 200–249). The five algorithms span the performance range observed in the 11-algorithm comparison: LAAVHA (lowest handover count), TOPSIS-Q (best MADM baseline), GRA (mid-range MADM), VIKOR (high handover count), and strongest-signal selection (single-attribute baseline). Three QoS metrics are evaluated: mean throughput (Mbps), mean delay (ms), and mean packet loss ratio (%). Table 4 and Figure 5 present the results.

Table 4: End-to-end QoS comparison across five representative algorithms (mean ± std, n=50, 50-second trajectories).

Algorithm	**Throughput (Mbps)**	**Delay (ms)**	**Packet Loss (%)**
LAAVHA	2.00 ± 0.12	1.00 ± 0.15	1.63 ± 0.18
TOPSIS-Q	1.98 ± 0.14	1.06 ± 0.18	1.63 ± 0.18
GRA	1.51 ± 0.22	1.39 ± 0.26	1.76 ± 0.24
Strongest-Signal	0.24 ± 0.08	2.29 ± 0.42	1.57 ± 0.20
VIKOR	1.78 ± 0.18	1.20 ± 0.21	1.62 ± 0.20

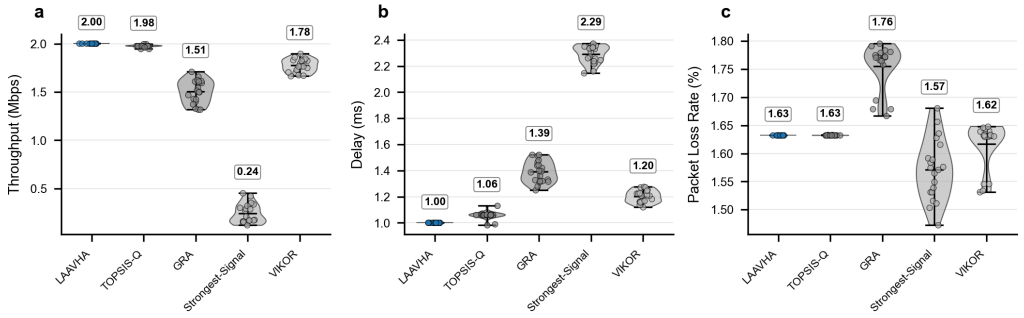


Figure 5: QoS metric distributions across five representative algorithms (n=50). (**a**) Throughput, (**b**) delay, (**c**) packet loss ratio. Violin plots show density; points denote individual runs, and horizontal bars mark the mean and range.

LAAVHA achieves the highest mean throughput (2.00 Mbps) and lowest mean delay (1.00 ms) while maintaining low packet loss (1.63%), comparable to TOPSIS-Q (1.63%) and VIKOR (1.62%). This demonstrates that LAAVHA's 46%–95% handover reduction compared to baselines (Figure 4) does not compromise service quality; rather, fewer handovers accompany superior connectivity. In contrast, strongest-signal selection—despite achieving

the lowest packet loss (1.57%)—suffers severely degraded throughput (0.24 Mbps, 88% lower than LAAVHA) and elevated delay (2.29 ms, 129% higher), illustrating that single-attribute optimization fails to maintain overall service quality. TOPSIS-Q achieves performance comparable to LAAVHA (throughput 1.98 Mbps, delay 1.06 ms), but at the cost of 85% more handovers (3.94 vs. 2.13), confirming that temporal prediction and adaptive weighting enable both stability and service continuity.

4.3. Comparison of Representative Decision Processes

To illustrate algorithm behaviors, Figure 6 compares LAAVHA, TOPSIS-Q, and Fuzzy-VHO under a 50-second trajectory (seed=200). TOPSIS-Q isolates the contribution of temporal prediction and dynamic weighting: it shares LAAVHA's TOPSIS framework but uses entropy weights and current-state-only ranking, providing a controlled comparison within the same decision paradigm. Fuzzy-VHO represents an alternative rule-based approach that reacts to local SINR and RSRP fluctuations through fuzzy membership functions, illustrating differences between adaptive multi-attribute fusion and rule-driven handover.

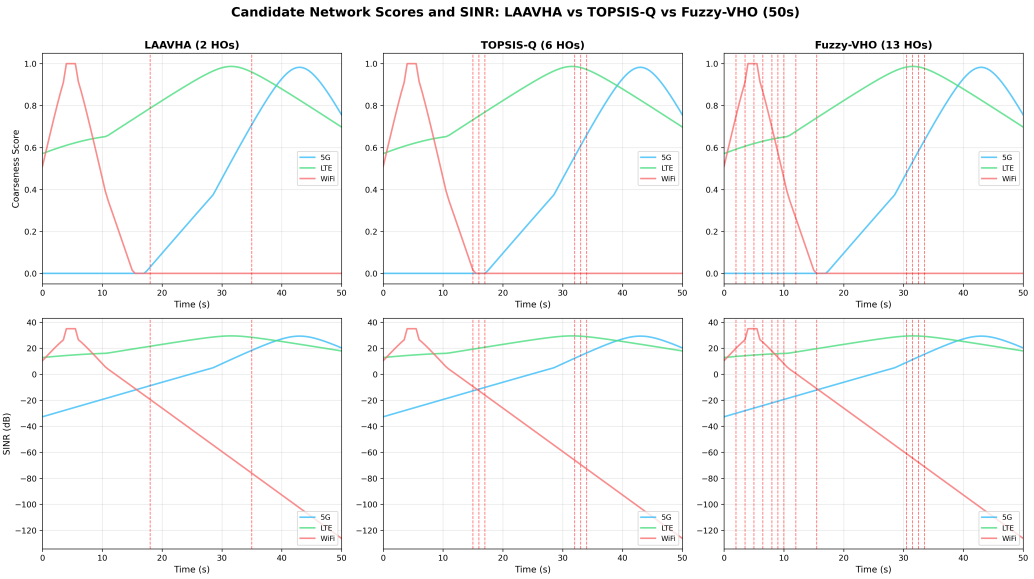


Figure 6: Candidate-network scores and SINR time series for LAAVHA, TOPSIS-Q, Fuzzy-VHO under a 50-second representative trajectory (seed=200) traversing the coverage boundaries in Figure 1. Vertical lines indicate handovers.

LAAVHA maintains stable decisions with 2 handovers corresponding to necessary network transitions as the UAV crosses major coverage boundaries (5G to Wi-Fi, then to LTE). Dual hysteresis and temporal prediction suppress reactive reassignments caused by transient fluctuations within each coverage region. Attention-based dynamic weights enable 5G selection during most of the mission despite not always having the highest SINR, by incorporating delay, throughput, and packet loss alongside signal strength. In contrast, TOPSIS-Q triggers 6 handovers due to repeated closeness-score convergence near coverage boundaries, while Fuzzy-VHO produces 13 handovers responding aggressively to local SINR fluctuations. These results confirm that temporal prediction and adaptive weighting stabilize handover decisions across extended trajectories.

4.4. Ablation-Study Validation

Ablation experiments quantify the contributions of LSTM prediction and attention-based dynamic weighting to LAAVHA. LAAVHA-L removes LSTM prediction by setting the predicted state equal to the current state while retaining attention weighting, improved TOPSIS, and dual hysteresis, testing temporal prediction. LAAVHA-A removes attention-based dynamic weighting, retains LSTM prediction, improved TOPSIS, and dual hysteresis, and replaces attention weights with entropy weights, testing dynamic-weight learning. Both variants and full LAAVHA use the same 250 random seeds (200−449) and 50-second simulation parameters. Figure 7 presents handover counts with scatter-overlaid bars, where each point represents one run, bar height is the mean, and error bars indicate ±1 standard deviation. LAAVHA achieves 2.13 (std=0.34), reflecting stable cooperation between prediction and dynamic weighting in extended trajectories. LAAVHA-L increases to 5.55 (std=0.64); removing temporal prediction produces more reactive decisions and 160% more handovers. LAAVHA-A reaches 24.25 (std=1.35); entropy weighting cannot adapt criteria to the flight phase and leads to frequent handovers. The full algorithm achieves an 11.4-fold reduction (2.13 vs. 24.25) compared to the no-attention variant. Dynamic weight adaptation is essential for handover stability.

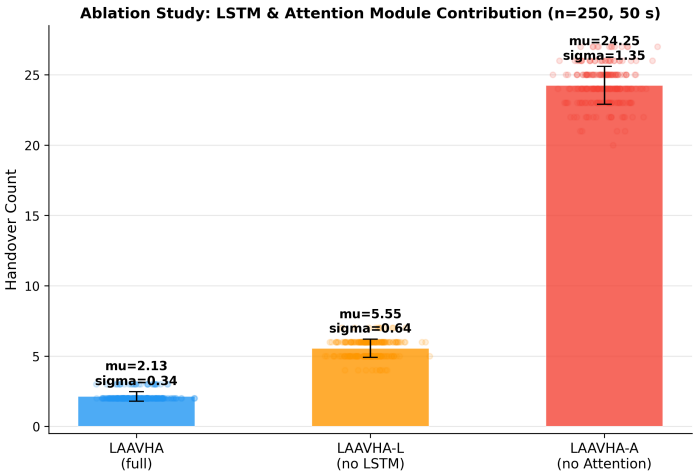


Figure 7: Handover-count comparison in the LAAVHA ablation study (mean ± std, n=250, 50-second trajectories).

The ablation results reveal strong synergy between LSTM prediction and attention-based dynamic weighting; neither module alone maintains low handover frequency in extended trajectories. Removing attention has the largest effect, increasing the mean from 2.13 to 24.25, an 11.4-fold rise. Flight- and network-state-based weights adapt to phase-specific requirements far better than static entropy weights. Removing the LSTM increases the mean to 5.55; temporal prediction contributes an additional 61% reduction (5.55→2.13) beyond dynamic weighting alone. Together, LSTM prediction provides temporal smoothing, while attention-based weighting prevents the ping-pong effect caused by fixed weights near coverage boundaries.

4.5. Scenario Adaptation and Enhancement Validation

To evaluate ALERA under rapid coverage-boundary crossings and abrupt channel changes, a competitive scenario is constructed with network A, analogous to 5G, and network B, analogous to Wi-Fi, having closeness scores near 0.50; A is initially serving. The scenario represents two data transmission conditions: timely migration under sustained degradation and handover suppression under transient disturbances from short obstructions or multipath effects. Figure 8 shares the time axis across four panels, with $t = 5 - 8$ s and $t = 12 - 15$ s denoting the same intervals in panels a–d. Light-blue Zone 1 represents sustained degradation, where A declines and B improves, requiring a correct A-to-B handover. Light-red Zone 2 represents transient

fluctuations, where sinusoidal variation and Gaussian noise are injected into B; short peaks provide no sustained mean advantage and should not trigger handover.

Panel (a) of Figure 8 shows raw closeness curves. The vertical axis represents the candidate's integrated score before risk penalization, with higher values indicating greater proximity to the ideal candidate. Network B maintains a persistent advantage in Zone 1, whereas its Zone 2 increase is a transient high-frequency oscillation.

Panel (b) shows handovers under the fixed low threshold ($\Delta_{\text{th}} = 0.03$); the vertical axis is raw closeness, and red dashed lines mark handover times and directions. Correct A→B and B→A handovers occur at $t = 6.6$ s and $t = 11.9$ s, respectively, corresponding to sustained degradation and subsequent recovery. In Zone 2, transient peaks trigger A→B at $t = 14.8$ s and B→A at $t = 15.1$ s, producing a false handover and subsequent ping-pong handover.

Panel (c) presents ALERA's risk-adjusted scores, with the vertical axis $C_i^{\text{robust}} = C_i - \lambda\sigma_i^C$, with green dashed lines indicating handovers. ALERA retains both necessary handovers while suppressing Zone 2 fluctuations and avoiding handover.

Panel (d) shows the adaptive threshold Δ_{th}, representing the required closeness advantage. In Zone 2, it increases from 0.03 to approximately 0.08 and gradually decreases after fluctuations subside. ADH raises the handover barrier only under unstable conditions.

The enhancement provides two key benefits. Under sustained degradation, it completes the necessary handover from degraded network A to better network B, avoiding excessive conservatism. Under transient variation, it suppresses false handovers caused by short peaks and prevents A→B→A ping-pong. RS-TOPSIS reduces the ranking advantage of volatile candidates, determining which network to switch to, while ADH raises the trigger threshold during fluctuations, determining when to switch. Together, they preserve necessary handovers, reduce false handovers, and limit associated link-reestablishment overhead.

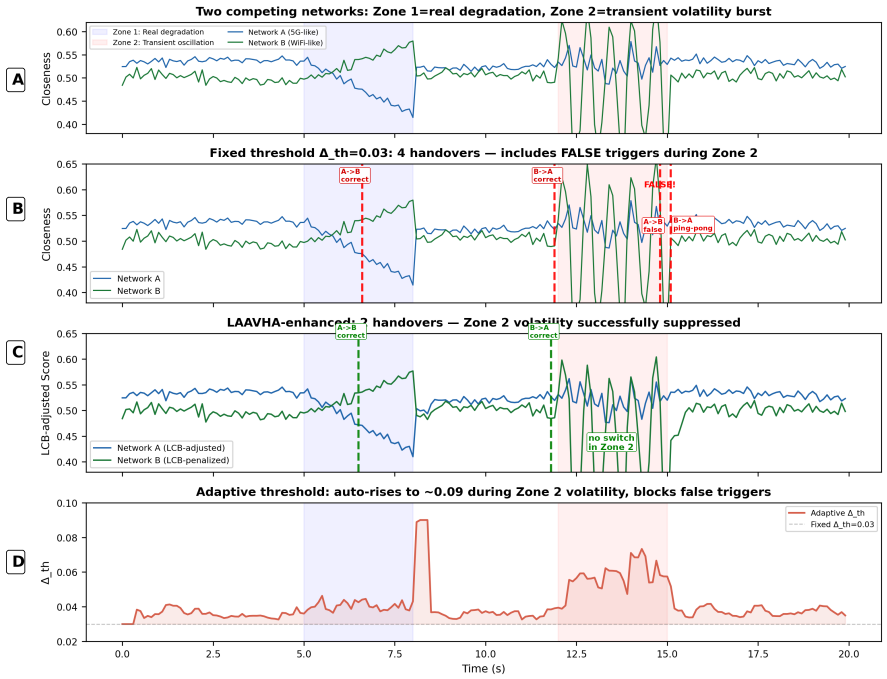


Figure 8: Adaptive-hysteresis validation of fixed low threshold ($\Delta_{\text{th}} = 0.03$) and ALERA under sustained degradation and transient fluctuation.

5. Conclusion

For VHO challenges arising from wide-area flight, heterogeneous coverage, and rapid link-state changes in UAV heterogeneous wireless networks, this paper proposes LAAVHA and its enhanced version, ALERA. LAAVHA integrates stacked-LSTM short-term state prediction, attention-based dynamic attribute weighting, improved TOPSIS using current and predicted states, and dual-hysteresis decision-making. This joint use of temporal evolution, flight state, and multi-attribute service requirements reduces handovers caused by transient boundary fluctuations. LAAVHA is further enhanced with adaptive hysteresis (ADH) and risk-sensitive TOPSIS (RS-TOPSIS) to form ALERA. ADH adapts the handover threshold and confirmation window to serving-network SINR variation and flight speed, while RS-TOPSIS penalizes historically volatile candidates, improving adaptation in both handover timing and network selection. Across 250 extended 50-second trajectories, LAAVHA achieves the lowest mean handover count among 11 algorithms (2.13), reducing handovers by 46.0% – 94.8% versus eight baselines while

maintaining the highest throughput and lowest delay among evaluated algorithms. Removing LSTM prediction or dynamic attention weighting increases the mean to 5.55 and 24.25, respectively. The two modules play complementary roles in handover stability. Scenario-adaptation experiments demonstrate that ALERA preserves necessary handovers under sustained degradation while suppressing false handovers caused by transient variation, confirming the effectiveness of adaptive decision parameters for UAV aerial missions in heterogeneous wireless networks.

CRediT authorship contribution statement

Hanming Sun: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Tong Liu**: Conceptualization, Supervision, Writing – review & editing, Funding acquisition. **Wen Su**: Software, Validation, Writing – review & editing. **Jie Lin**: Formal analysis, Writing – review & editing. **Junsong Luo**: Investigation, Writing – review & editing. **Yu Tian**: Resources, Writing – review & editing. **Wenshu Sun**: Data curation, Writing – review & editing. **Bin Duo**: Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The simulation code and trained model weights supporting the findings of this study are available from the corresponding author upon reasonable request. Raw simulation data cannot be publicly deposited due to the large file size of ns-3 trajectory outputs.

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